cipherDNS: Encrypted DNS-over-HTTPS Resolver

A Privacy-Preserving Local Caching Proxy with Automated Zeek Traffic Verification

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1 Abstract / Summary

Traditional Domain Name System (DNS) queries operate over unencrypted UDP on port 53. Because these queries are transmitted in cleartext, any entity observing the local link—including malicious actors on public Wi-Fi, compromised gateway routers, and Internet Service Providers (ISPs)—can passively monitor browsing activities, reconstruct complete web traffic profiles, or actively execute DNS spoofing and cache poisoning attacks. This project presents **cipherDNS**, a high-performance local DNS caching resolver and privacy proxy that intercepts standard operating system and browser DNS queries over UDP port 53 and forwards them to upstream resolvers using cryptographically authenticated **DNS-over-HTTPS (DoH, RFC 8484)**.

Key Implemented Features:

- **Standard UDP Port 53 Ingestion:** Seamlessly interoperates with OS resolvers, systemd-resolved, and client applications (e.g., Google Chrome, Firefox, curl, dig) without requiring client modifications.
- **RFC 8484 DoH Protocol Client:** Encapsulates binary wire-format DNS queries in HTTP/2 POST requests with strict TLS 1.3 certificate validation using the Mozilla Root CA bundle.
- **Multi-Provider Support:** Provides instant runtime switching between Cloudflare (1.1.1.1), Google Public DNS (8.8.8.8), Quad9 (9.9.9.9), and AdGuard.
- **Dynamic TTL-Aware In-Memory Cache:** Implements RFC 1035/2181 standards-compliant caching. Answer records have their TTL values dynamically decremented based on elapsed cache time. Expired records are automatically evicted, and cache hits are delivered with sub-millisecond latency (<1 ms).
- **Real-Time Dark-Themed Web Dashboard:** Features a prominent status ring showing live query states ('cipherDNS running', 'Making a DoH request to Cloudflare...', 'Successfully delivered cached entry'), an interactive TTL cache table ordered with most recent entries at the top, a manual query tester, and a live event stream via WebSockets.
- **Automated Zeek Network Privacy Verification:** Integrates with Zeek (/opt/zeek/bin/zeek) to capture and analyze network pcaps, scientifically proving that traditional DNS leaks domain queries in dns.log while cipherDNS conceals all traffic inside TLS 1.3 encrypted application records.

What Works in the Final Prototype: The prototype is fully operational. System DNS queries generated by desktop browsers for arbitrary sites (e.g., netflix.com) are intercepted locally, resolved over DoH, cached with decremented TTLs, displayed in real time on the web dashboard, and verified as private via automated Zeek packet inspection.

2 Problem Definition, Requirements & Threat Model

2.1 Problem Scope & Assumptions: We assume the client device host is uncompromised, but the local network (Wi-Fi hotspot, LAN switches, upstream routers, ISP) is untrusted and subject to eavesdropping. Standard client applications expect RFC 1035 UDP DNS on port 53. Our objective is to secure the transmission channel between the local client and upstream authoritative/recursive resolvers.

2.2 Functional Requirements:

- **FR-1:** Accept inbound standard DNS datagrams over UDP port 53 (or unprivileged fallback port 5300).
- **FR-2:** Parse RFC 1035 wire-format queries, extracting Question Name, Type, and Class.
- **FR-3:** Query in-memory cache before initiating external network requests.
- **FR-4:** For cache misses, forward query via RFC 8484 DoH using HTTP/2 POST with application/dns-message.
- **FR-5:** Enforce dynamic TTL decrementing on cache hits and match client query Transaction IDs.
- **FR-6:** Provide an interactive web dashboard for real-time monitoring and administrative control.

2.3 Security Requirements & Threat Model:

Threat Vector / Adversary	Attack Mechanism	cipherDNS Mitigation Strategy	Status
Passive Eavesdropper (ISP / Rogue Wi-Fi AP)	Sniffing plaintext UDP port 53 packets to harvest user browsing history.	TLS 1.3 encryption over TCP port 443; zero DNS packets transmitted across the WAN.	**PROTECTED**
Active Man-in-the-Middle (DNS Spoofing / Hijacking)	Injecting forged UDP DNS responses with malicious IP addresses.	Strict X.509 certificate chain validation prevents spoofed upstream responses.	**PROTECTED**
Cache Poisoning Attack	Flooding rogue records to corrupt resolver cache.	Only cryptographically verified DoH responses from authenticated CAs are cached.	**PROTECTED**
SNI Leakage (TLS Client Hello)	Observer inspects Server Name Indication during subsequent HTTPS connection.	Out-of-scope for DNS layer; requires Encrypted Client Hello (ECH) at browser layer.	**OUT OF SCOPE**
Compromised Local Host	Rootkit/malware reading loopback traffic on 127.0.0.1.	Out-of-scope; host-level security is orthogonal to network proxy encryption.	**OUT OF SCOPE**

3 Architecture and Design

cipherDNS is designed as an asynchronous, non-blocking pipeline composed of three primary subsystems: the **UDP Ingestion Engine**, the **TTL Cache Engine**, and the **DoH Upstream Dispatcher**, coupled with a reactive **FastAPI Web Dashboard** and **Zeek Packet Inspector**.

3.1 End-to-End Resolution Data Flow:

1. Client OS or browser (e.g. Chrome) dispatches standard UDP query for netflix.com (A record) to 127.0.0.1:53.
2. DNSServer captures the datagram via asyncio.DatagramProtocol and parses the binary wire format using dnspython.
3. The cache is queried using normalized key ('netflix.com', A, IN). If a valid, unexpired entry exists (Cache Hit), the response TTL is decremented to the remaining lifetime, the query Transaction ID is stamped, and the packet is immediately dispatched back to the client (<1 ms).

4. On a Cache Miss, an event is broadcast to the web dashboard (updating the status ring to 'Making a DoH request to Cloudflare...').
5. The query is forwarded over an established HTTP/2 TLS 1.3 connection to https://cloudflare-dns.com/dns-query with header Content-Type: application/dns-message.
6. Upon receiving HTTP 200, the response is parsed, verified, stored in the cache with the upstream minimum TTL, returned to the client, and broadcast to the dashboard.

3.2 Security Checks and Error Handling:

- **Strict TLS Certificate Verification:** Prior to dispatching queries, the upstream TLS certificate is verified using trusted Mozilla root CAs. Hostname checking (check_hostname=True) and ssl.CERT_REQUIRED are enforced.
- **Upstream Timeout & Failure Resilience:** If the upstream DoH endpoint times out (threshold 3.5s) or returns an HTTP error code (e.g. 500, 502), cipherDNS synthesizes an RFC-compliant SERVFAIL (RCODE 2) packet preserving the client's query ID, preventing client hangs.
- **Malformed Packet Sanitization:** Corrupted or truncated datagrams trigger graceful parsing exceptions without terminating the server loop.

4 Implementation

The prototype is implemented in **Python 3.13** utilizing asynchronous I/O (asyncio) for high-concurrency DNS handling. The table below details component responsibilities, libraries, and attribution:

Module Path	Key Responsibilities & Important Algorithms	Libraries Used	Author / Source
`cipherdns.core.cache`	Thread-safe in-memory store; dynamic remaining TTL calculation; RFC response wire rewriting with remaining TTL; LRU eviction.	threading, dnspython	Custom Written
`cipherdns.core.doh_client`	RFC 8484 HTTP/2 client; strict TLS verification; cryptographic certificate evidence extraction; SERVFAIL synthesizer.	httpx[http2], ssl, certifi	Custom Written
`cipherdns.core.plaintext_client`	Standard unencrypted UDP port 53 resolver client used for side-by-side comparative testing.	socket, dnspython	Custom Written
`cipherdns.core.server`	Async UDP DNS server; query dispatching; cache coordination; unprivileged port fallback handling.	asyncio, dnspython	Custom Written
`cipherdns.analysis.zeek_analyzer`	PCAP generation; automated execution of /opt/zeek/bin/zeek; TSV log parsing for dns.log leakage.	scapy, subprocess	Custom Written
`cipherdns.web.app`	FastAPI REST endpoints and WebSocket event broadcaster streaming real-time status and cache table updates.	fastapi, uvicorn, jinja2	Custom Written

Implementation Challenges Overcome:

1. *Dynamic TTL Decrementing:* Naive DNS caching returns the static initial TTL, causing downstream clients to reset

cache expirations. We implemented dynamic wire-format record modification where msg.answer[i].ttl is rewritten to max(1, remaining_ttl) before sending.
2. *Privileged Port 53 on Linux:* Binding port 53 requires root privileges. We architected the server with an automatic fallback to port 5300 when unprivileged, while providing seamless resolvectl and run.sh scripts for system-wide port 53 operation.
3. *Campus Network Firewall Filtering:* During live testing, we discovered the local campus network filters port 443 connections to Google (8.8.8.8) and Quad9 (9.9.9.9). Our robust timeout mechanism intercepted this cleanly and demonstrated graceful SERVFAIL generation without application crashing.

5 Testing, Results, and Security Evaluation

We performed extensive testing across three categories: functional unit/integration tests, adversarial security tests, and empirical network traffic analysis using Zeek.

Test Case ID & Description	Test Type	Expected Outcome	Actual Outcome	Result
TC-01: DoH Resolution over RFC 8484 (Cloudflare)	Functional	NOERROR response with valid A records; latency recorded.	Resolved netflix.com -> 207.45.72.1 in 25ms.	**PASS**
TC-02: Cache Hit & Dynamic Decrement	Functional	Sub-millisecond response; TTL decremented by elapsed seconds.	Delivered in 0.8ms; TTL decremented from 14s to 9s.	**PASS**
TC-03: Cache Expiration & Pruning	Functional	Expired entry evicted; triggers fresh upstream query.	Entry purged after 14s; new DoH query dispatched.	**PASS**
TC-04: Upstream TLS Certificate Audit	Security	Valid certificate chain, trusted root CA, TLS 1.3 cipher.	Validated: TLS_AES_256_GCM_SHA384 via Mozilla Root CA.	**PASS**
TC-05: Upstream Timeout Resilience	Negative / Failure	Catches timeout after 3.5s; returns RFC SERVFAIL.	Intercepted timeout; returned clean SERVFAIL (RCODE 2).	**PASS**
TC-06: Corrupted Response Handling	Adversarial	Prevents parser crash; returns SERVFAIL.	Caught wire exception; synthesized SERVFAIL.	**PASS**
TC-07: Zeek Plaintext Traffic Leakage	Network Audit	Zeek records cleartext query in dns.log.	Domain netflix.com logged in plain sight.	**VERIFIED**
TC-08: Zeek cipherDNS Privacy Audit	Network Audit	Zero records in dns.log; TLS 1.3 on port 443.	dns.log is completely EMPTY; 100% encrypted.	**VERIFIED**

5.2 Zeek Network Traffic Evidence Analysis:

To scientifically evaluate the privacy improvement, we analyzed identical requests for netflix.com under Traditional DNS vs cipherDNS DoH using Zeek 9.0.0. The parsed log outputs are shown below:

[Traditional Plaintext DNS] Zeek dns.log Extract:

```
1793279467.87   192.168.1.105:53821 -> 8.8.8.8:53   udp   query: netflix.com   A   IN
Vulnerability: The full domain name 'netflix.com' is visible to any passive eavesdropper on the local
network link.
```

[cipherDNS Encrypted DoH] Zeek dns.log & conn.log Extract:

```
dns.log: [EMPTY - ZERO QUERIES RECORDED]
conn.log: 1793279467.87   192.168.1.105:48210 -> 1.1.1.1:443   tcp   service: ssl (TLS 1.3 Encrypted Application Data)
Privacy Evaluation: Query contents are completely concealed inside TLS 1.3 records. Network observers only see HTTPS packets to Cloudflare.
```

6 Limitations and Lessons Learned

6.1 Achieved Requirements: The prototype successfully achieves encrypted DNS-over-HTTPS resolution, RFC-compliant dynamic TTL caching, real-time web telemetry with circular animation states, and multi-provider selection.

6.2 Known Weaknesses and Limitations:

- **Bootstrap DNS Dependency:** If DoH provider endpoints are specified by hostname (e.g., cloudflare-dns.com), the resolver requires initial bootstrap resolution to obtain the provider's IP address. cipherDNS mitigates this by maintaining hardcoded bootstrap IPs (1.1.1.1, 8.8.8.8).
- **Upstream Provider Centralization:** While DoH protects against local eavesdroppers and ISPs, the chosen upstream provider (Cloudflare or Google) observes user queries. Users must trust the provider's privacy policy.
- **SNI Leakage in Browser Connections:** DoH encrypts the DNS resolution, but subsequent HTTPS connections may leak the domain in the TLS Client Hello SNI header unless the browser supports Encrypted Client Hello (ECH).
- **Latency Characteristics:** Initial DoH queries introduce a modest overhead (20–40ms) compared to plaintext UDP due to TLS handshakes. However, local caching completely amortizes this, reducing repeat query latency to under 1ms.

7 Deployment and Future Work

7.1 Setup and Execution Instructions:

1. **Install Dependencies:** `pip install -r requirements.txt`
2. **Run with Standard Port 53:** `sudo python3 run.py --dns-port 53 --web-port 8000`
3. **Run Unprivileged (Demo Mode):** `python3 run.py --dns-port 5300 --web-port 8000`
4. **Redirect System DNS:** `./scripts/setup_system_dns.sh`
5. **Access Web Dashboard:** Navigate to http://localhost:8000 in any web browser.
6. **Execute Test Query:** `dig @127.0.0.1 -p 53 netflix.com`

7.2 Future Work Roadmap:

- **DNSSEC Validation:** Incorporate cryptographic signature verification (RRSIG/DNSKEY) to detect upstream tampering.
- **DNS-over-QUIC (DoQ, RFC 9250):** Implement DoQ over UDP port 853 to eliminate head-of-line blocking and reduce connection establishment latency.
- **Oblivious DoH (ODoH, RFC 9230):** Decouple client identity from DNS queries via an oblivious proxy relay, eliminating trust in the upstream resolver.

8 Appendix: Project-Specific Minimum Technical Evidence Index

This section documents the formal technical evidence required by the CS416 Encrypted DNS Resolver project specification.

8.1 Evidence of DNS Query Completing Through DoH (RFC 8484):

The trace below captures an end-to-end DNS resolution for netflix.com executed through Cloudflare DoH:

```
HTTP/2 Request Headers:
:method = POST
:scheme = https
:authority = cloudflare-dns.com
:path = /dns-query
content-type = application/dns-message
accept = application/dns-message
content-length = 29 bytes [RFC 1035 wire query for netflix.com IN A]

HTTP/2 Response:
:status = 200 OK
content-type = application/dns-message
cf-ray = a5226941c8b9613f-BOM
Payload: 72 bytes [ID: 12870, RCODE: NOERROR, Answers: 207.45.72.1, 207.45.73.1, TTL: 14s]
```

8.2 Evidence of Upstream Certificate Validation:

Cryptographic audit extracted via Python ssl and certifi root CA validation:

Audit Parameter	Cryptographic Verification Value
Upstream Hostname	`cloudflare-dns.com (Port 443)`
TLS Protocol Version	**TLSv1.3**
Negotiated Cipher Suite	**TLS_AES_256_GCM_SHA384** (256-bit symmetric key)
CA Validation Status	**VALID_TRUSTED** (against Mozilla Root CA bundle)
Issuer Common Name	SSL.com SSL Intermediate CA ECC R2
Subject Alternative Names	`cloudflare-dns.com, *.cloudflare-dns.com, one.one.one.one`
Certificate Serial Number	`4ED03304C46B87A8C2EB5569DB9EBA0C`
Validity Window	Dec 31 19:20:01 2025 GMT – Dec 21 19:20:01 2026 GMT

8.3 Evidence of Handling Invalid Responses, Upstream Failures & Timeouts:

1. **Invalid / Corrupted Upstream Payload:** When tested against corrupted wire bytes (b'NOT_A_VALID_DNS_WIRE_MESSAGE_HEADER'), cipherDNS intercepted the parsing exception and returned an RFC SERVFAIL (RCODE 2) response preserving the client's query ID.
2. **Upstream Timeout Handling:** When routed to an unresponsive blackhole address (192.0.2.1), cipherDNS triggered its 3.5-second timeout, caught httpx.TimeoutException, and dispatched a clean SERVFAIL to the client.
3. **Firewall Network Failure:** In network environments where connections to Google (8.8.8.8:443) are filtered, cipherDNS caught httpx.ConnectError and gracefully returned SERVFAIL without terminating server execution.